

# Manas Arora

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## EDUCATION

**Maharaja Surajmal Institute of Technology, Delhi**  
Bachelor of Technology in Computer Science and Engineering

Aug 2023 – July 2027  
CGPA: 8.82/10

**Dewan Public School, Meerut**  
Class XII  
Class X

Mar 2023  
Percentage: 91.2%  
Percentage: 92.6%

## EXPERIENCE

**CG Infinity**  
Software Development Engineer (SDE) Intern

June 2026 – Present  
Noida, India

- Developed full-stack solutions using **Next.js** and **.NET**, working across frontend interfaces, backend APIs, and enterprise application workflows.
- Built and integrated enterprise functionality using **Salesforce Lightning Web Components (LWC)** and **Apex**, gaining hands-on experience with Salesforce application development.
- Worked with **AWS S3**, **AWS Lambda**, and **AWS Glue** for cloud-based storage, serverless processing, and data integration workflows.
- Gained practical experience with **API integration**, **cloud services**, **backend development**, and **enterprise application architecture** across multiple technology stacks.

## PROJECTS

### NexChat – Real-Time Messaging Platform

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- Engineered a full-stack real-time messaging platform using MongoDB, Express.js, React.js, Node.js, and Socket.IO, supporting one-to-one messaging, typing indicators, online/offline presence, and database-backed read receipts.
- Re-architected WebSocket event broadcasting from direct socket IDs to Socket.IO rooms, improving read-receipt reliability across multiple tabs and socket reconnections.
- Optimized real-time interactions using debounced global user search and typing-indicator events to reduce redundant API calls and socket traffic.
- Containerized the backend using Docker and deployed the application on Render with a separate static frontend deployment.

### YouTube RAG – AI Video Search Platform

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- Architected a three-service RAG platform consisting of a React frontend, Node.js orchestration layer, and FastAPI Python microservice for semantic search and conversational interaction with YouTube content.
- Separated authentication/session management from the retrieval and generation pipeline, establishing clear service responsibilities across the application.
- Integrated LangChain, Pinecone, NVIDIA NIM, vector embeddings, and semantic retrieval to generate context-aware responses from YouTube transcripts.
- Implemented non-blocking ingestion with frontend polling to handle vector-database readiness and deployment cold-start constraints without paid infrastructure.

### Quirk – Social Media Platform

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- Built a full-stack social media platform using MongoDB, Express.js, React.js, and Node.js with support for posts, likes, follows, and activity feeds.
- Developed responsive user interfaces using React.js, Tailwind CSS, and reusable component-based architecture.
- Designed and implemented RESTful backend APIs with Express.js and MongoDB for user profiles, posts, social relationships, and activity data.
- Implemented optimistic UI updates for user interactions to improve perceived responsiveness and reduce unnecessary waiting for server responses.

## TECHNICAL SKILLS

|                            |                                                                                      |
|----------------------------|--------------------------------------------------------------------------------------|
| <b>Languages:</b>          | JavaScript, TypeScript, C++, Python, C#, SQL                                         |
| <b>Frontend:</b>           | React.js, Next.js, HTML, CSS, Tailwind CSS, Redux Toolkit                            |
| <b>Backend:</b>            | Node.js, Express.js, FastAPI, .NET, REST APIs, JWT Authentication, Socket.IO         |
| <b>Cloud &amp; Data:</b>   | AWS S3, AWS Lambda, AWS Glue, PostgreSQL, MongoDB, Pinecone, Prisma                  |
| <b>Salesforce:</b>         | Lightning Web Components (LWC), Apex                                                 |
| <b>AI / Generative AI:</b> | LangChain, LangGraph, RAG, Vector Embeddings, Semantic Search, NVIDIA NIM            |
| <b>Tools:</b>              | Git, GitHub, Docker, Postman, Vercel, Render, MongoDB Atlas                          |
| <b>Core Concepts:</b>      | Data Structures & Algorithms, OOP, DBMS, Operating Systems, REST APIs, System Design |

## ACHIEVEMENTS

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- Secured **8.73/10 CGPA** in B.Tech Computer Science and Engineering at MSIT Delhi.
- Built and independently deployed a production-grade three-service RAG platform integrating Node.js, FastAPI, LangChain, NVIDIA NIM, and Pinecone.
- Designed and deployed a real-time messaging platform with Socket.IO, Docker, database-backed read receipts, and room-based WebSocket broadcasting.